

<Title of the Work>

<Subtitle>

Bachelor Thesis

Bachelor Course on Creative Computing
at St. Pölten University of Applied Sciences

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Declaration of Honour

I hereby declare that

- I have written the work at hand on my own without help from others and I have used no other resources and tools than the ones acknowledged.
- I have complied with the Standards of good scientific practice in accordance with the St. Pölten UAS' Guidelines for Scientific Work when writing this work.
- I have neither published nor submitted the work at hand to another higher education institution for assessment or in any other form as examination work.

Regarding the use of generative artificial intelligence tools such as chatbots, image generators, programming applications, paraphrasing and translation tools, I declare that

- ☒ no generative artificial intelligence tools were used in the course of this work.
- ☐ I have used generative artificial intelligence tools to proof-read this work.
- ☐ I have used generative artificial intelligence tools to create parts of the content of this work. I certify that I have cited the original source of any generated content. The generative artificial intelligence tools that I used are acknowledged at the respective positions in the text.

Having read and understood the St. Pölten UAS' Guidelines for Scientific Work, I am aware of the consequences of a dishonest declaration.

Abstract

Introduction: Warum behandeln wir das Thema

Purpose: Welches Problem soll gelöst werden

Method: Wie wurde die Problemlösung gemacht

Product: Was war das Ergebnis

Conclusion: Was sind die Folgerungen / Schlussfolgerungen aus den gewonnen Erkenntnissen

keine Referenzen und Zitate

Kurzfassung

Das Abstract auf deutsch.

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1 Introduction

1.1 Problem Statement

This thesis is about the utilisation of voice commands for player control in video games. The goal of this paper is to verify if the player can successfully navigate in a sample video game environment using spoken natural language instead of traditional input methods, such as keyboard, mouse or controller. These methods rely on hands for input and control, making it impossible for handicapped people who lack the motor skills and/or can't use their hands to take part in the activity. While games that support voice commands do exist, they rely on preexisting voice commands. This poses a challenge by requiring the user to remember the exact commands that need to be said to trigger actions. The system in this paper will utilise a language model to translate natural language into specific game actions, thereby eliminating the need for command memorisation.

Putting above in more technical terms with the current AI transcribing abilities it's possible to detect and transcribe human voice in near real time. That natural language transcription would be interpreted by an LLM with direct access to the game engine. Such deep integration could produce an input method that's more flexible than traditional voice commands for navigation.

This paper will focus on a game environment similar to a pen and paper RPG game, since players tend to have a variety of options available to them at any time, promoting non-linear thinking and ambiguous commands. This would allow for deep exploration into interpreting natural language as game commands.

The technical challenge for this paper is: creating a natural voice input pipeline that can interpret and execute player commands in the provided gaming environment with a high accuracy. This involves addressing software, usability, and hardware challenges.

1.2 Research questions/objectives

How would the participants describe the experience of using voice input in the presented game environment?

How do users feel about using the voice modality compared to traditional input?

1.3 Research Strategy

Given the personal nature of this topic, and the limitations on time and resources, this thesis will be focusing on employing qualitative methods. This allows for empathising with the users on an individual level to understand the nuance of their experiences with the voice input method.

The main qualitative method used is going to be guided interviews with the subjects. That will seek out the emotions and insights from the subjects.

This strategy, relying on qualitative will allow for reaching results that will be a combination of voices of all participants.

1.4 Scope and limitations

1.5 Thesis structure overview

1. Introduction
2. Literature Review
3. Methodology
4. Results
5. Discussion
6. Conclusion
7. Appendix

2 Literature Review

This is the literature review

3 Methodology

This chapter is about my Methodology

4 Results / Ergebnisse

Presenting found literature in a useful way

4.1 First Section

Ich bin Text, Text, Text¹

4.1.1 First Subsection

¹<http://mfg.fhstp.ac.at>

5 Discussion / Diskussion

Comparison of presented technologies/methods/projects

Kritische Diskussion / Vergleich der Ansätze

Welche Methoden werden zumeist genutzt, warum?

Überblick / Zusammenfassung der gefundenen Literatur in einer sinnvollen Kategorisierung
/ Charakterisierung

6 Conclusion / Fazit

Was kann man daraus lernen?

Was fehlt?

Ideen für zukünftige Forschung

7 Example

!!! Please delete this chapter after finishing your work !!!

7.1 Settings

To add your name and the title of your work, please use the “Settings.tex” file! Additionally, switch there between German and English version.

7.2 How to Make Sections and Subsections

Use section and subsection commands to organize your document. $\text{\LaTeX}$ handles all the formatting and numbering automatically. Use `ref` and `label` commands for cross-references.

7.2.1 How to Make Lists

You can make lists with automatic numbering ...

1. Like this,
2. and like this.

... or bullet points ...

- Like this,
- and like this.

... or with words and descriptions ...

Word Definition

Concept Explanation

Idea Text

7.3 Section

You have to write text between each headline.

7.4 Citation

This part describes the three types of citations which are possible. Therefore, two different methods are possible the

- “autocite” version (Stoiber et al., 2023)
- “textcite” version Tjoa et al. (2022)

7.5 Direct Citation

The maximum for a direct citation is a 1/2 page.

Overview first, zoom and filter, then details-on-demand (Shneiderman, 1996)

7.6 Floating Text Citation

Shneiderman (1996) defined the Visual Information Seeking Mantra as “Overview first, zoom and filter, then details-on-demand”.

7.7 Indirect Citation

Some text which summarizes a paper or a book chapter. This could take several lines. Find attached a citation of a website (Kaley, 2018, p. 12).

7.8 Figures

To place a figure use the following code example

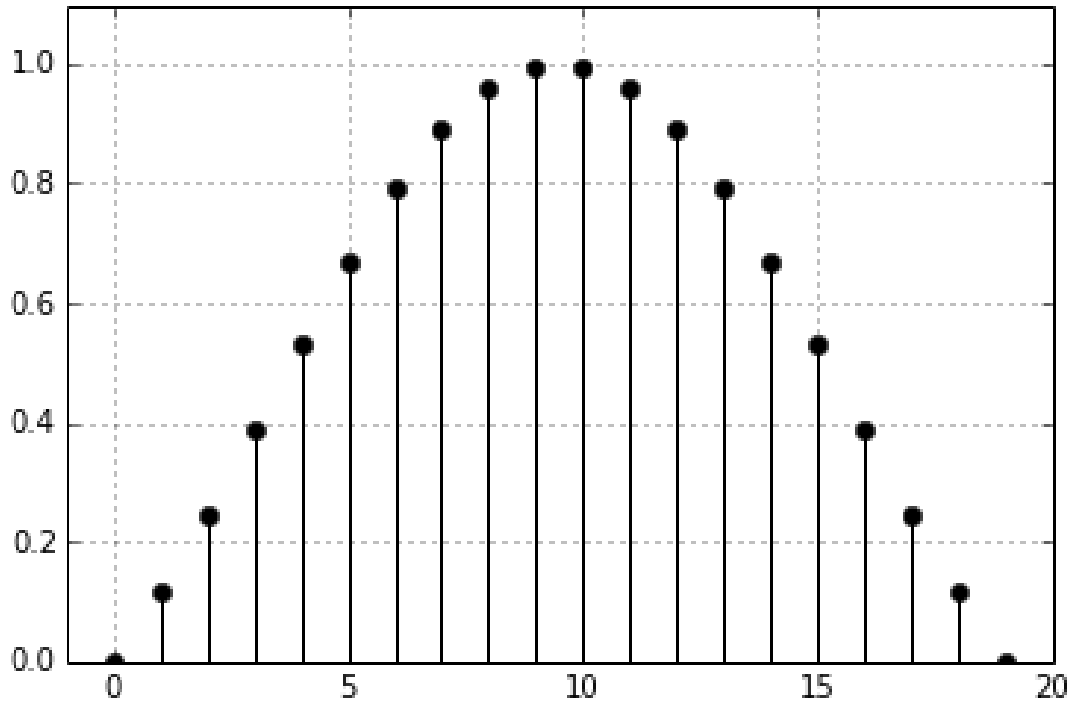


Figure 7.1. Interactive data exploration with multiple devices.

$$\int a, b$$

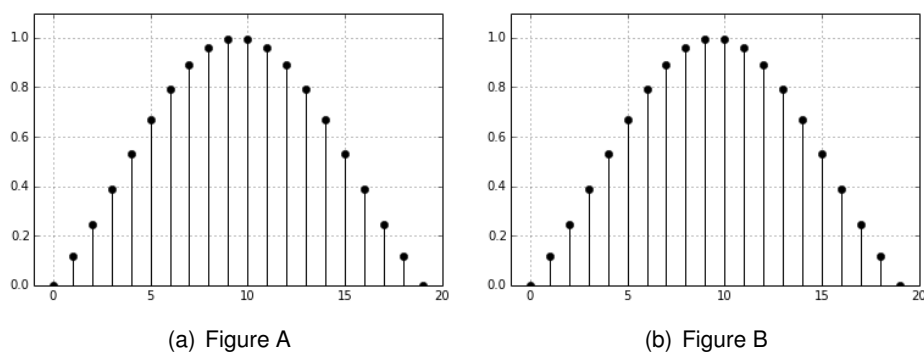


Figure 7.2. Wearables worn for experiments 1, 2, and 3.

Refer to a figure in the following forms:

If you take a look at Figure 7.1 ...

... text text (see Figure 7.1) ...

7.9 Listings

Listing 7.1. A bit of source code.

```
1 if( true == questions )
2 {
3     std::cout << "Let_me_google_it_for_you";
4 }
5 else
6 {
7     std::cout << "Great";
8 }
```

Now lets take a look at Listing 7.1.

7.10 Table

Table 7.1. My caption with a very useful description. die kann auch etwas länger sein und über mehrere Zeilen gehen und so weiter.

Item		
Animal	Description	Price (\$)
Gnat	per gram	13.65
	each	0.01
Gnu	stuffed	92.50
Emu	stuffed	33.33
Armadillo	frozen	8.99

For the fast generation of tables from Excel use <http://www.heise.de/download/excel2latex.html>

7.11 Equations

$\text{\LaTeX}$ is great at typesetting equations. Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed random variables with $E[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2 < \infty$, and let

$$S_n = \frac{X_1 + X_2 + \dots + X_n}{n}$$

This was a equation without a label.

$$S_n = \frac{1}{n} \sum_i^n X_i \tag{7.1}$$

This is the reference to equation 7.1.

denote their mean. Then as n approaches infinity, the random variables $\sqrt{n}(S_n - \mu)$ converge in distribution to a normal $\mathcal{N}(0, \sigma^2)$.

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Listings

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Appendices

A Appendix

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B Appendix

LoHrem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum. LoHrem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum. LoHrem